



BobSim

Ramp-Steer Vehicle Characterization

Measured- a_y ramp-response isolines with robust metric fits

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Run Notes

- Open-loop constant-rate handwheel ramp at the configured test velocities
- steerStart = 2.0 s; trim all pre-ramp samples
- Measured a_y is the analysis axis
- Metrics describe the ramp-response path, not fully settled equilibrium holds
- $|a_y| < 1.0 \text{ m/s}^2$ is excluded; linear band = 1.0 to 4.0 m/s^2
- Ramp terminates when local lateral gain drops 20% from the linear reference
- Limit metrics are local derivatives at the largest measured $|a_y|$ reached
- Tire-lift protection terminates if any tire falls below the configured normal-load threshold
- Raw plots are rendered; summary metrics use robust smoothed derivatives

RampSteerEval Summary 12.5 m/s

Metric	Value	Units
Measured a_y Range	1.00 → 13.86	m/s^2
Linear Sideslip Gradient	0.00239	$\frac{\text{rad}}{\text{m/s}^2}$
	1.34	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	0.00016	$\frac{\text{rad}}{\text{m/s}^2}$
	0.09	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00093	$\frac{\text{rad}}{\text{m/s}^2}$
	0.52	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	0.00241	$\frac{\text{rad}}{\text{m/s}^2}$
	1.36	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.04462	$\frac{\text{rad}}{\text{m/s}^2}$
	25.08	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.05358	$\frac{\text{rad}}{\text{m/s}^2}$
	30.11	$\frac{\text{deg}}{g}$
Linear Roll Gradient	1.431	$\frac{\text{deg}}{g}$
Limit Roll Gradient	1.566	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-17.4	$\text{N} \cdot \text{m}$
Peak Torque a_y	13.86	m/s^2

RampSteerEval Summary 15.0 m/s

Metric	Value	Units
Measured a_y Range	1.00 → 12.37	m/s^2
Linear Sideslip Gradient	0.00119	$\frac{\text{rad}}{\text{m/s}^2}$
	0.67	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	−0.00042	$\frac{\text{rad}}{\text{m/s}^2}$
	−0.24	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00126	$\frac{\text{rad}}{\text{m/s}^2}$
	0.71	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	0.00247	$\frac{\text{rad}}{\text{m/s}^2}$
	1.39	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.03372	$\frac{\text{rad}}{\text{m/s}^2}$
	18.96	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.04013	$\frac{\text{rad}}{\text{m/s}^2}$
	22.56	$\frac{\text{deg}}{g}$
Linear Roll Gradient	1.431	$\frac{\text{deg}}{g}$
Limit Roll Gradient	1.533	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-15.4	$\text{N} \cdot \text{m}$
Peak Torque a_y	12.37	m/s^2

RampSteerEval Summary 17.5 m/s

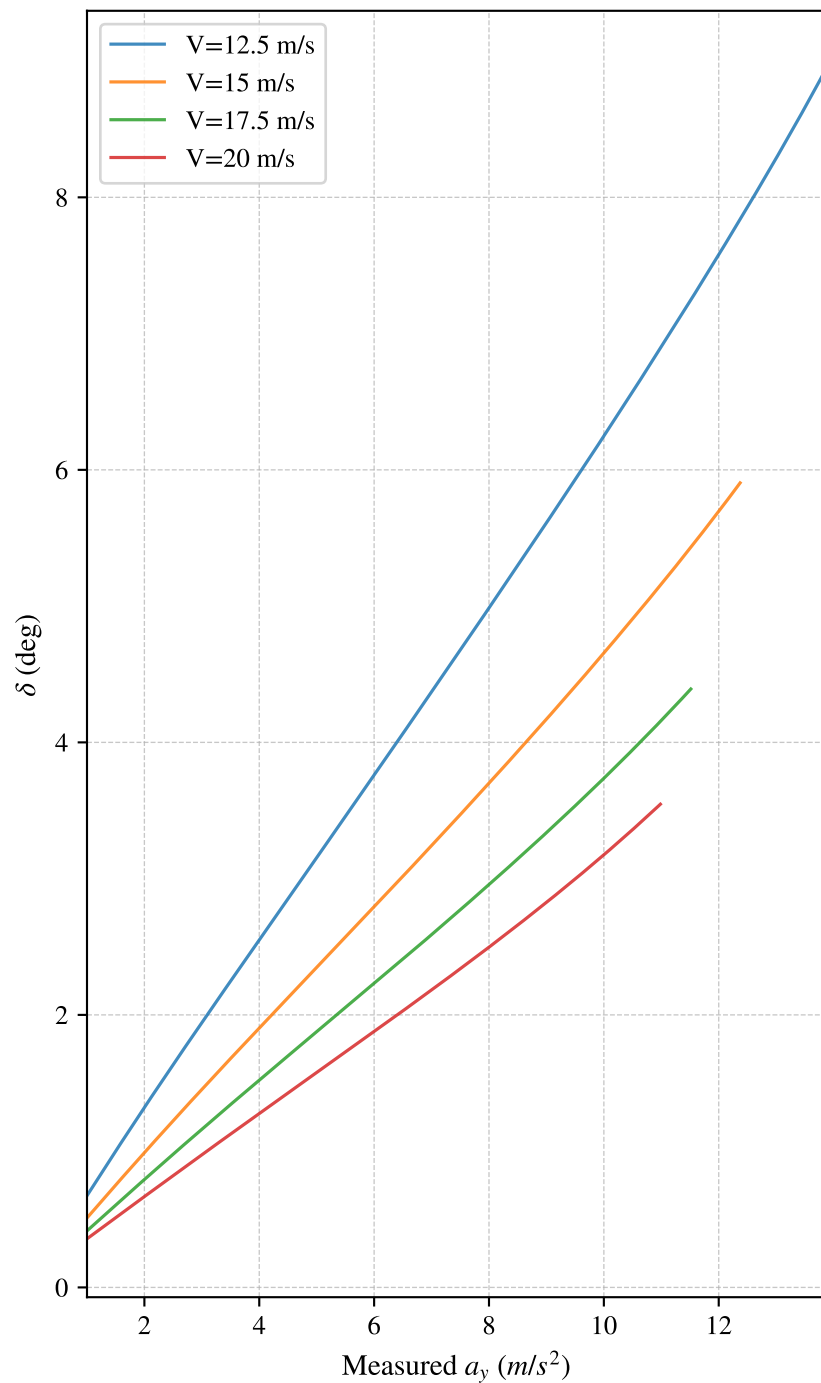
Metric	Value	Units
Measured a_y Range	1.00 → 11.51	m/s^2
Linear Sideslip Gradient	0.00039	$\frac{\text{rad}}{\text{m/s}^2}$
	0.22	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	−0.00087	$\frac{\text{rad}}{\text{m/s}^2}$
	−0.49	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00136	$\frac{\text{rad}}{\text{m/s}^2}$
	0.76	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	0.00258	$\frac{\text{rad}}{\text{m/s}^2}$
	1.45	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.02667	$\frac{\text{rad}}{\text{m/s}^2}$
	14.99	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.03230	$\frac{\text{rad}}{\text{m/s}^2}$
	18.15	$\frac{\text{deg}}{g}$
Linear Roll Gradient	1.436	$\frac{\text{deg}}{g}$
Limit Roll Gradient	1.527	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-14.2	$\text{N} \cdot \text{m}$
Peak Torque a_y	11.51	m/s^2

RampSteerEval Summary 20.0 m/s

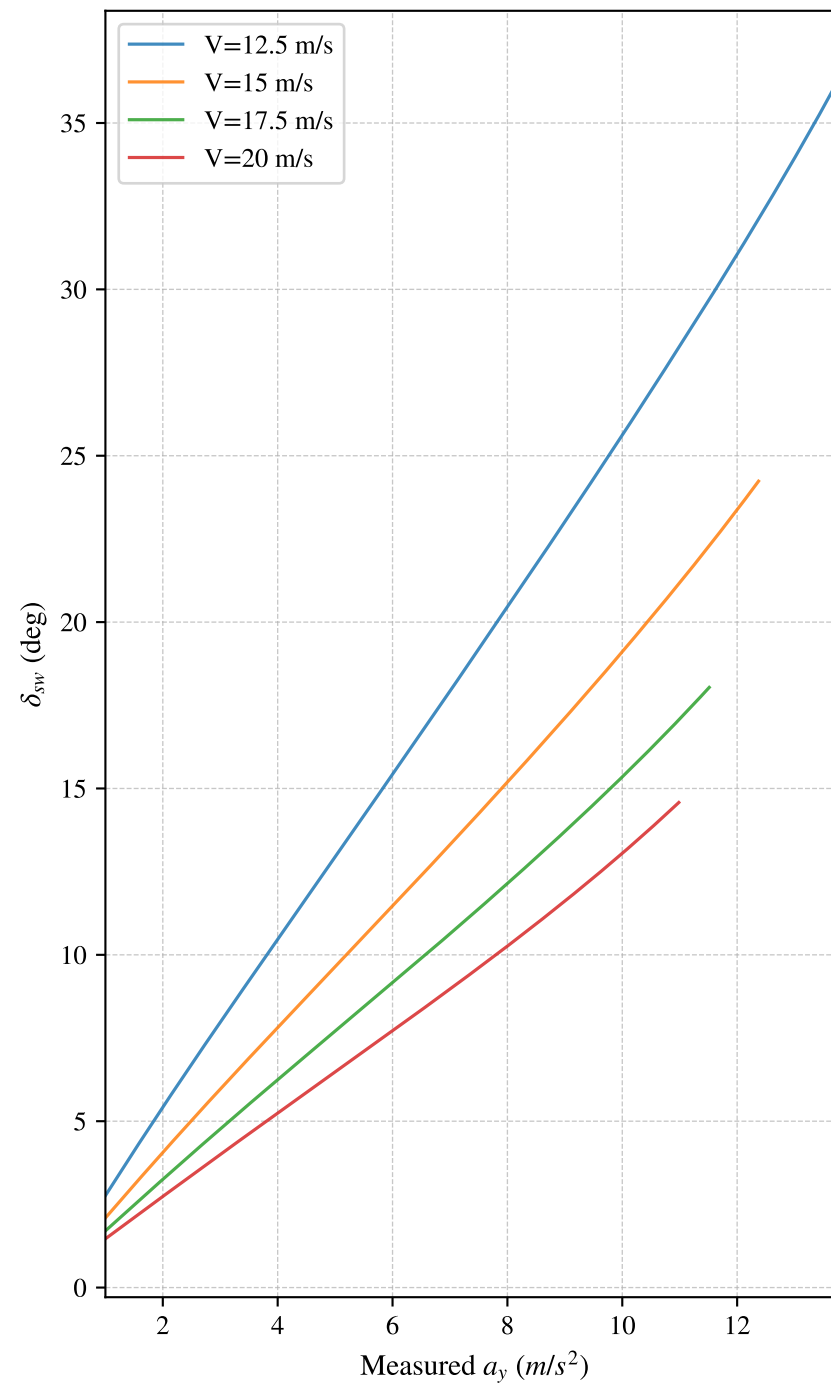
Metric	Value	Units
Measured a_y Range	1.00 → 10.98	m/s^2
Linear Sideslip Gradient	-0.00008	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.04	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	-0.00116	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.65	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00148	$\frac{\text{rad}}{\text{m/s}^2}$
	0.83	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	0.00268	$\frac{\text{rad}}{\text{m/s}^2}$
	1.51	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.02233	$\frac{\text{rad}}{\text{m/s}^2}$
	12.55	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.02746	$\frac{\text{rad}}{\text{m/s}^2}$
	15.43	$\frac{\text{deg}}{g}$
Linear Roll Gradient	1.444	$\frac{\text{deg}}{g}$
Limit Roll Gradient	1.536	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-13.6	$\text{N} \cdot \text{m}$
Peak Torque a_y	10.98	m/s^2

Raw Steering Samples vs Measured a_y

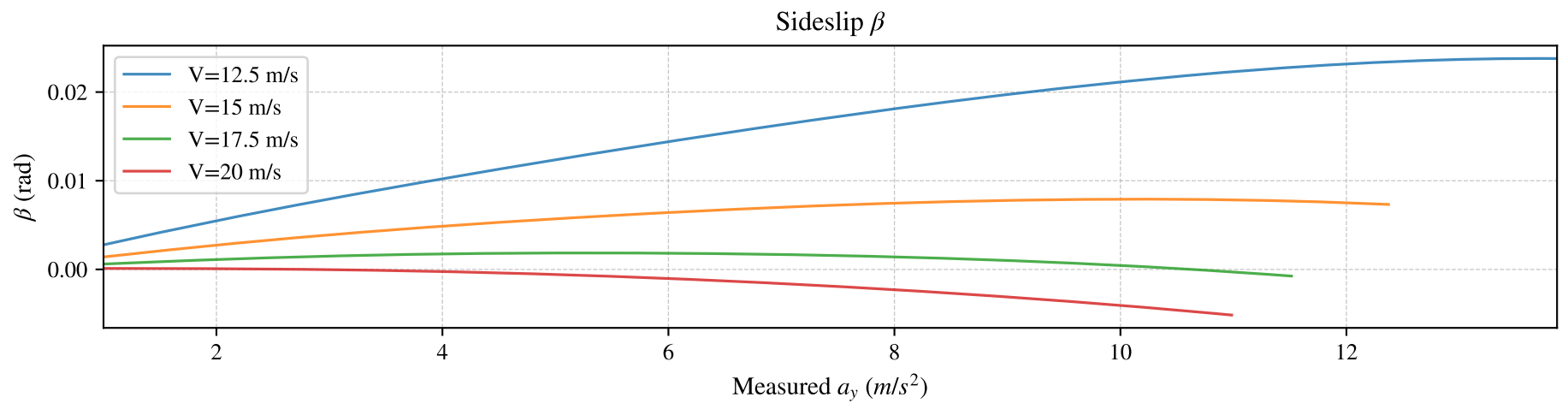
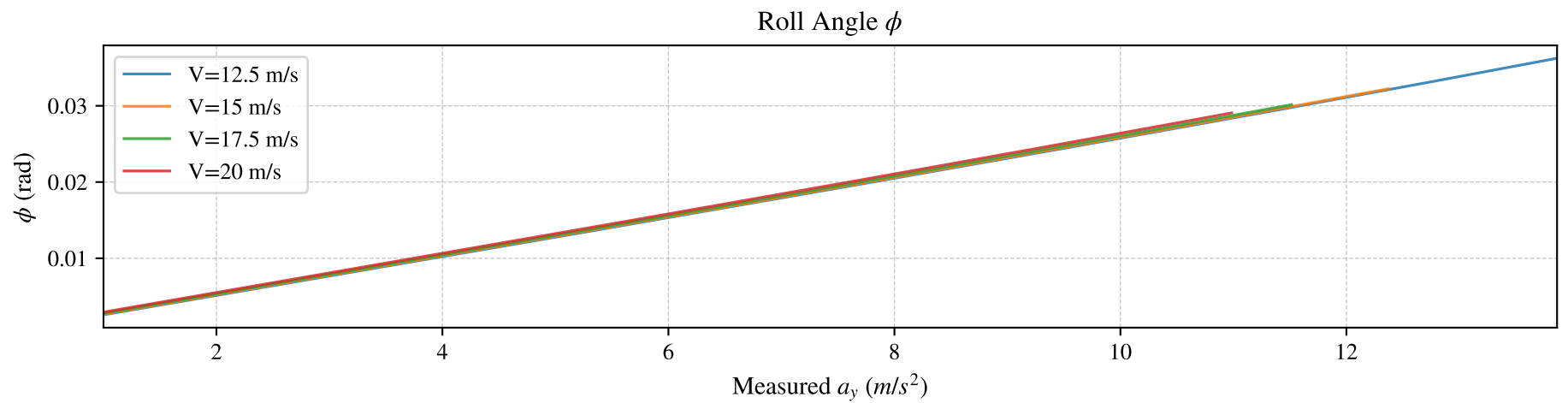
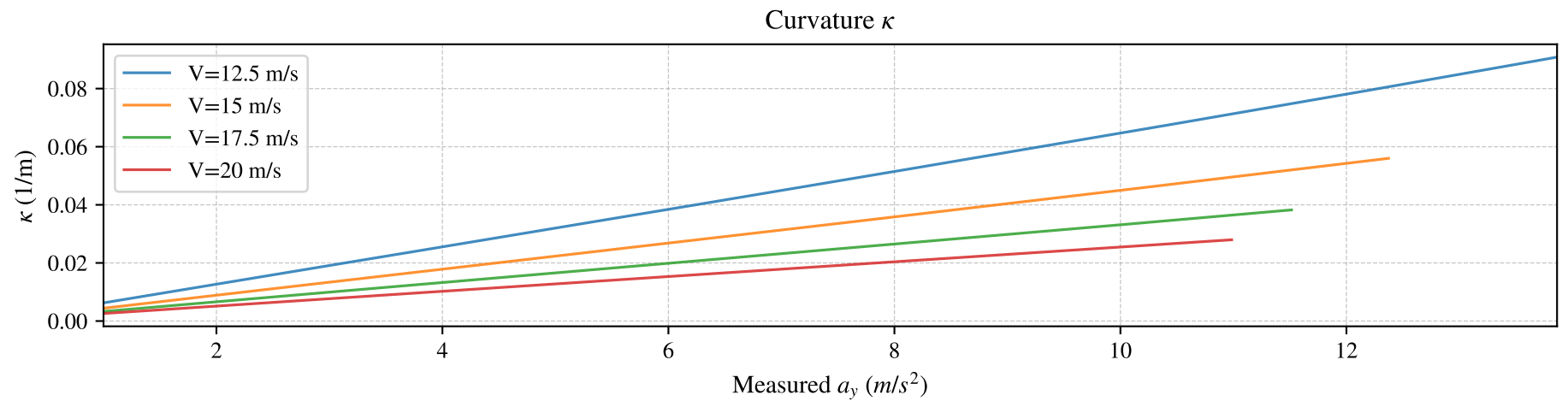
Roadwheel Angle δ



Handwheel Angle δ_{sw}



Raw Vehicle State Samples vs Measured a_y



Raw Steering Effort Samples vs Measured a_y

